

A fourfold max-convolution inequality on three-point integer supports

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Abstract

Let F consist of three distinct integers, let g, k be supported on F and positive there, and let f be a nonzero finitely supported nonnegative function on $\mathbb{Z}$. For max-convolution $(f \star g)(s) = \max_{x+y=s} f(x)g(y)$ and reflection $\widetilde{h}(x) = h(-x)$, we prove

$$\|f \star \widetilde{gk}\|_1^4 < \|f \star g\|_1^4 \|k \star k \star k \star k\|_1.$$

There is no restriction on the support size or positive values of f . For non-arithmetic triples, the proof combines a uniform reflected-quotient bound with a classification of fourfold collisions. For arithmetic progressions, scalar polynomial estimates and piecewise-linear potentials cover all positive weight profiles. The potentials telescope over finite sequences, and strictness follows from a scalar margin or an endpoint term.

Keywords. max-convolution; sumset; difference set; three-point support.

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1 Statement and notation

For nonnegative finitely supported functions on $\mathbb{Z}$, define max-convolution and reflection by

$$(h_1 \star h_2)(s) = \max_{x+y=s} h_1(x)h_2(y), \quad \widetilde{h}(x) = h(-x).$$

An empty maximum is zero, and $\|h\|_1 = \sum_x h(x)$. Functions specified on a finite set are extended by zero outside it. Products such as gk are pointwise products. In particular, if F is the common support of g and k , write

$$Q = \|f \star \widetilde{gk}\|_1 = \sum_x \max_{p \in F} f(x+p)g(p)k(p), \tag{1}$$

$$P = \|f \star g\|_1 = \sum_s \max_{p \in F} f(s-p)g(p), \tag{2}$$

$$T = \|k \star k \star k \star k\|_1 = \sum_s \max_{\substack{p_1, \dots, p_4 \in F \\ p_1 + \dots + p_4 = s}} \prod_{j=1}^4 k(p_j). \tag{3}$$

Each output in (3) contributes its largest product once; ordered representations are not added.

*Developed with AI research agents. The manuscript gives algebraic proofs covering every parameter range in the stated theorem.

Theorem 1. *Let $F \subset \mathbb{Z}$ have exactly three elements. For every $g, k : F \rightarrow (0, \infty)$ and every nonzero finitely supported $f : \mathbb{Z} \rightarrow [0, \infty)$,*

$$Q^4 < P^4 T. \quad (4)$$

The max-convolution operation and its use in cardinality inequalities are part of the framework of Matolcsi, Ruzsa, Shakan and Zhelezov [2, Definition 8.1 and Section 8]. Gyarmati, Hennecart and Ruzsa [1, equation (9), p. 177] asked for which integers h the inequality

$$|A - B| \leq |A + B| |hB|^{1/h}$$

holds for all finite nonempty sets in an abelian group; their discussion on p. 182 left the cases $h = 4, 5$ unresolved. Here hB is the set of sums of h elements of B . Taking $g = k = 1_F$ and $f = 1_A$ in Theorem 1 gives the strict fourfold inequality when $A \subset \mathbb{Z}$ is finite and nonempty and $|F| = 3$. The theorem concerns the displayed weighted inequality on three support points; the assertion for arbitrary finite supports is a separate question.

Related weighted work includes the Prékopa–Leindler chapter of Green, Matolcsi, Ruzsa, Shakan and Zhelezov [3]. A comparison with that chapter and with the journal version of [2] remains incomplete. We make no priority or optimality claim for the theorem or its methods. All mathematical ingredients used in the proof are supplied below.

The proof first bounds Q/P uniformly in the geometry of a triple. Section 4 completes the non-arithmetic case. Sections 5–8 treat progressions by the position and order of the g weights. Positivity gives $P > 0$ throughout, so it suffices to prove $Q < T^{1/4}P$.

2 A uniform reflected-quotient bound

Proposition 2. *For every three-point integer support, positive g , nonnegative k , and nonzero finite nonnegative f , let $M = \max_{p \in F} k(p)$ and $S = \sum_{p \in F} k(p)$. Then*

$$Q \leq \left(M + \frac{S - M}{4} \right) P. \quad (5)$$

For two support points the corresponding bound is $Q \leq (\max k + \min k/4)P$.

Proof. First take $k = 1_F$. Translate a point where g is largest to zero and divide g by that maximum. Write the other points as p, q , with weights $c, d \in (0, 1]$. Either offset may be negative. Put $W = \sum_x f(x)$, $C = c + d$, and $D_0 = c^2 + d^2$. For nonnegative x, y, z ,

$$\max(x, cy, dz) \leq x + c(y - cx)_+ + d(z - dx)_+. \quad (6)$$

The right side is at least x . If $y \leq cx$, then $cy \leq x$; if $y \geq cx$, its first two terms are $(1 - c^2)x + cy \geq cy$. The dz branch follows in the same way.

Let

$$P_p = \sum_s \max(f(s), cf(s - p)), \quad P_q = \sum_s \max(f(s), df(s - q)).$$

The change of variable $s = x + p$ gives the orientation-sensitive identity

$$\sum_x (f(x + p) - cf(x))_+ = P_p - cW.$$

Since $P_p, P_q \leq P$, summing (6) yields

$$Q(1_F) \leq CP + (1 - D_0)W. \quad (7)$$

Also $Q(1_F) \leq (1 + C)W$ and $W \leq P$. If $D_0 \leq 1$, these inequalities imply

$$\frac{Q(1_F)}{P} \leq 1 + C - D_0 = 1 + c(1 - c) + d(1 - d) \leq \frac{3}{2}.$$

If $D_0 > 1$, substitute $W \geq Q(1_F)/(1 + C)$ into the negative W term in (7). Because $D_0 \geq C^2/2$ and $C \leq 2$,

$$\frac{Q(1_F)}{P} \leq \frac{C(1 + C)}{C + D_0} \leq \frac{2(1 + C)}{2 + C} \leq \frac{3}{2}.$$

The case $D_0 = 1$ was included in the first argument.

For a pair $I \subset F$, use the same argument after normalizing its larger g weight to one and omitting the d term. With P_I denoting its denominator, it gives

$$Q(1_I) \leq (1 + c - c^2)P_I \leq \frac{5}{4}P_I \leq \frac{5}{4}P.$$

For a single point j , $Q(1_{\{j\}}) = g(j)W \leq P$. These normalizations multiply the relevant numerator and denominator equally, so the comparisons hold for the original weights.

For fixed f, g , the function $Q(k)$ is positively homogeneous and subadditive, as a sum of maxima of linear functions of k . Order the three weights as $M \geq m \geq n \geq 0$, at points a, b, c . The identity

$$k = (M - m)1_{\{a\}} + (m - n)1_{\{a, b\}} + n1_F$$

has nonnegative coefficients. Consequently

$$\frac{Q(k)}{P} \leq (M - m) + \frac{5}{4}(m - n) + \frac{3}{2}n = M + \frac{m + n}{4}.$$

The two-point version follows from the same decomposition without the last term. Ties and zero k weights are included directly. $\square$

3 Scalar estimates for the fourfold total

Order the support as $p_0 < p_1 < p_2$ and normalize its k weights to $(1, u, v)$, where $u, v > 0$. Throughout this section $r = T^{1/4}$ and

$$\begin{aligned} S_4(t) &= 1 + t + t^2 + t^3 + t^4, \\ L(u, v) &= 1 + u + u^2 + u^3 + u^4 + u^3v + u^2v^2 + uv^3 + v^4, \\ E(u, v) &= 1 + v + v^2 + v^3 + v^4 + u(1 + v + v^2 + v^3). \end{aligned}$$

Lemma 3. *For every ordered three-point support,*

$$T \geq L(u, v), \quad T \geq E(u, v), \quad T \geq S_4(u), \quad T \geq S_4(v), \quad r > \max(1, u, v).$$

On an arithmetic progression,

$$T = \begin{cases} L(u, v), & u^2 \geq v, \\ E(u, v), & u^2 \leq v. \end{cases} \quad (8)$$

Proof. The five fourfold outputs using p_0, p_1 end at $4p_1$; those using p_1, p_2 start there. Their only common position is $4p_1$, and their nine products sum to L . Put $\delta = p_2 - p_0$ and $\epsilon = p_1 - p_0$, so $0 < \epsilon < \delta$. The five positions $4p_0 + j\delta$, $0 \leq j \leq 4$, and the four positions $4p_0 + \epsilon + j\delta$, $0 \leq j \leq 3$, strictly interlace. Their products are v^j and uv^j , proving the bound by E . Restricting to either pair containing p_0 gives the two S_4 bounds. The three distinct diagonal outputs give $T \geq 1 + u^4 + v^4$.

For a progression, translate and scale the points to $0, 1, 2$. A count pair (n_1, n_2) with output $s = n_1 + 2n_2$ has product $u^s(v/u^2)^{n_2}$. Choosing the smallest or largest allowable n_2 according to $v \leq u^2$ or $v \geq u^2$ gives (8). $\square$

Lemma 4. For every ordered three-point support and all $u, v > 0$,

$$r > \max(1, t) + \frac{1}{4} \min(1, t) \quad (t = u, v), \quad (9)$$

$$r > u + \frac{1}{4}(1 + v), \quad (10)$$

$$r > 1 + \frac{1}{4}u + \frac{5}{32}v. \quad (11)$$

Proof. The first assertion follows from $T \geq S_4(t)$ and the identities

$$\begin{aligned} S_4(t) - (1 + t/4)^4 &= \frac{5}{8}t^2 + \frac{15}{16}t^3 + \frac{255}{256}t^4, \\ S_4(t) - (t + 1/4)^4 &= \frac{5}{8}t^2 + \frac{15}{16}t + \frac{255}{256}. \end{aligned}$$

For (10), put $J(v) = 5(v - 1)^2 + 4v > 0$. Direct subtraction gives

$$L - \left(u + \frac{1+v}{4}\right)^4 = \frac{J(v)u^2}{8} + \frac{3(1+v)J(v)u}{16} + 1 + v^4 - \frac{(1+v)^4}{256}.$$

The last term is at least $31(1+v^4)/32 > 0$, since $(1+v)^4 \leq 8(1+v^4)$.

For (11), set $c = 5/32$. If $v \leq u^2$, expand $L - (1 + u/4 + cv)^4$. In the negative terms use

$$v \leq u^2, \quad uv \leq u^3, \quad v^2, u^2v \leq u^4, \quad uv^2 \leq u^3v, \quad v^3 \leq u^2v^2.$$

Before substituting c , the resulting lower bound is

$$\begin{aligned} &(5/8 - 4c)u^2 + (15/16 - 3c)u^3 + (255/256 - 6c^2 - 3c/4)u^4 \\ &+ (1 - c/16 - 3c^2)u^3v + (1 - 3c^2/8 - 4c^3)u^2v^2 + (1 - c^3)uv^3 + (1 - c^4)v^4. \end{aligned}$$

At $c = 5/32$ this is

$$\frac{15}{32}u^3 + \frac{375}{512}u^4 + \frac{939}{1024}u^3v + \frac{999}{1024}u^2v^2 + \frac{32643}{32768}uv^3 + \frac{1047951}{1048576}v^4 > 0. \quad (12)$$

If $u^2 \leq v$, hold v fixed and define $H(u) = E(u, v) - (1 + u/4 + cv)^4$ on $[0, \sqrt{v}]$. Then $H''(u) = -\frac{3}{4}(1 + u/4 + cv)^2 < 0$ and

$$H(0) > S_4(v) - (1 + v/4)^4 > 0.$$

At $t = \sqrt{v}$, the exact value $H(t)$ is (12) with $(u, v) = (t, t^2)$: both equal $\sum_{j=0}^8 t^j - (1 + t/4 + ct^2)^4$. It is positive. Concavity places H above the chord joining these positive endpoint values, completing the proof. $\square$

Lemma 5. For every ordered support,

$$r > \frac{8}{11}(1 + u) > \frac{2}{3}(1 + u), \quad r > u + \frac{1}{3} \quad (0 < u \leq 2). \quad (13)$$

On an arithmetic progression one also has

$$r^2 \geq u^2 + 2v, \quad \frac{v}{r} < r - u, \quad \frac{v}{r} < \frac{r}{2}. \quad (14)$$

Proof. For the first bound,

$$16S_4(u) - 5(1 + u)^4 = (u - 1)^2(11u^2 + 18u + 11) \geq 0,$$

and $5 \cdot 11^4 = 73205 > 65536 = 16 \cdot 8^4$. For $u \leq 2$,

$$\begin{aligned} S_4(u) - (u + 1/3)^4 &= -\frac{1}{3}u^3 + \frac{1}{3}u^2 + \frac{23}{27}u + \frac{80}{81} \\ &\geq -\frac{1}{3}u^2 + \frac{23}{27}u + \frac{80}{81} \geq \frac{5}{27}u + \frac{80}{81} > 0. \end{aligned}$$

Here $u^3 \leq 2u^2$ and $u^2 \leq 2u$ justify the comparisons.

For (14), pair progression output j with $8 - j$. When $u \geq \sqrt{v}$, the four pairs, in order $j = 0, 1, 2, 3$, are at least

$$2v^2, \quad 2uv^{3/2} \geq 2v^2, \quad 2u^2v, \quad 2u^3\sqrt{v} \geq 2u^2v;$$

the middle output is u^4 . When $u \leq \sqrt{v}$, pairs $j = 0, 2$ are each at least $2v^2$, pairs $j = 1, 3$ are each at least $2uv^{3/2} \geq 2u^2v$, and the middle output is $v^2 \geq u^4$. Thus $T \geq (u^2 + 2v)^2$ in both cases. It follows that $r^2 > 2v$ and

$$(r^2 - v)^2 - u^2r^2 = r^2(r^2 - u^2 - 2v) + v^2 > 0.$$

Since $r^2 - v > 0$, this gives $r^2 > ur + v$, proving the remaining bounds. $\square$

4 Non-arithmetic triples

Proposition 6. *If F is not an arithmetic progression, then*

$$T^{1/4} > M + \frac{S - M}{4}, \quad M = \max_F k, \quad S = \sum_F k.$$

Hence Theorem 1 holds for every non-arithmetic triple.

Proof. If a largest k weight is at the middle point, normalize the ordered weights to $(1, u, v)$. The result is (10). Otherwise reflect and scale so the weights are $(1, u, v)$ with $0 < u, v \leq 1$. Translate the support and divide its positive coordinates by their greatest common divisor to obtain

$$F' = \{0, p, q\}, \quad 0 < p < q, \quad \gcd(p, q) = 1.$$

These transformations preserve all equalities of fourfold sums. A count pair (i, j) has product $u^i v^j$, where $i, j \geq 0$ and $i + j \leq 4$, and position $pi + qj$. Write

$$\mathcal{H}(u, v) = \sum_{\substack{i, j \geq 0 \\ i + j \leq 4}} u^i v^j.$$

It has 15 terms. If two count pairs collide, coprimality gives $\Delta i = qt$, $\Delta j = -pt$ for a nonzero integer t . Since $|\Delta i| \leq 4$, there are no collisions when $q \geq 5$. For $q = 3, 4$, only $|t| = 1$ is possible, and every collision class has two members. The full list is

Support F'	Fourfold total T
$\{0, 1, 3\}$	$\mathcal{H} - (1 + u + v) \min(u^3, v)$
$\{0, 2, 3\}$	$\mathcal{H} - (1 + u + v) \min(u^3, v^2)$
$\{0, 1, 4\}$	$\mathcal{H} - \min(u^4, v)$
$\{0, 3, 4\}$	$\mathcal{H} - \min(u^4, v^3)$
$\{0, p, q\}, q \geq 5$	$\mathcal{H}$

For $(p, q) = (1, 3)$, the three paired products are (u^3, v) , (u^4, uv) , and (u^3v, v^2) . For $(p, q) = (2, 3)$, they are (u^3, v^2) , (u^4, uv^2) , and (u^3v, v^3) . For $q = 4$ the only pair is (u^4, v^p) . The case $q = 2, p = 1$ is the excluded progression; $q = 4, p = 2$ is not coprime. This proves completeness of the table.

Every loss in the table is at most $(1 + u + v)u^3$. In particular the $q = 4$ loss is at most $u^4 \leq (1 + u + v)u^3$. The difference between this common lower bound and the target is

$$\begin{aligned} \mathcal{H} - (1 + u + v)u^3 - \left(1 + \frac{u + v}{4}\right)^4 \\ = \frac{5}{8}(u^2 + v^2) + \frac{1}{4}uv - \frac{1}{16}u^3 + \frac{13}{16}(u^2v + uv^2) + \frac{15}{16}v^3 \\ - \frac{1}{256}u^4 - \frac{1}{64}u^3v + \frac{125}{128}u^2v^2 + \frac{63}{64}uv^3 + \frac{255}{256}v^4. \end{aligned}$$

Since $u, v \leq 1$, the total magnitude of the three negative terms is at most $21u^2/256$, whereas the positive u^2 term is $160u^2/256$. The difference is therefore positive. This proves $r > 1 + (u + v)/4$, including ties in the largest k weight. Combine this strict bound with Proposition 2. $\square$

5 Progressions and local potentials

For the rest of the proof let F be an arithmetic progression. Translation, reflection, and positive scaling of g and k allow the ordered weights to be written

$$g = (1, a, b), \quad a > 0, \quad 0 < b \leq 1, \quad k = (1, u, v), \quad u, v > 0. \quad (15)$$

Thus the larger endpoint of g is first. A common factor in g multiplies Q, P equally; a common factor in k multiplies Q, r equally. Reflection is applied simultaneously to f, g , and k , and reindexes all three totals.

After translating the progression to $\{0, d, 2d\}$ with $d > 0$, split the sequence f into its residue classes modulo d . On each class, index consecutive entries by $i \in \mathbb{Z}$ and put

$$N(x, y, z) = \max(x, auy, bvez), \quad D(x, y, z) = \max(bx, ay, z). \quad (16)$$

The corresponding contributions to Q, P are exactly

$$Q = \sum_i N(f_i, f_{i+1}, f_{i+2}), \quad P = \sum_i D(f_i, f_{i+1}, f_{i+2}). \quad (17)$$

For P , this is the ordinary convolution evaluated at the index $i + 2$. Both sums include the zero entries before and after the support. The fourfold total depends only on u, v , as in (8). We prove the desired inequality on each nonzero residue class; adding over classes then preserves strictness.

Lemma 7. *Suppose $\Phi : [0, \infty)^2 \rightarrow \mathbb{R}$ satisfies $\Phi(0, 0) = 0$ and*

$$N(x, y, z) \leq rD(x, y, z) + \Phi(x, y) - \Phi(y, z) \quad (x, y, z \geq 0). \quad (18)$$

Then $Q \leq rP$ for every finite sequence. If (18) is strict at $(0, 0, z)$ for every $z > 0$, then $Q < rP$ for every nonzero finite sequence.

Proof. All but finitely many potential values vanish, so the sum of their differences is zero by an index shift. At the first positive entry of a nonzero sequence there is a triple $(0, 0, z)$ with $z > 0$. $\square$

One parameter range needs no potential. If $br \geq 1$, then

$$N \leq \max(1/b, u, bv)D \leq rD. \quad (19)$$

Since $r > v$ and $b \leq 1$, the padded triple $(0, 0, z)$ is strict. In particular this settles $b = 1$. Below we may assume $br < 1$.

The endpoint cases will use the following elementary verification rule.

Lemma 8. *Let $0 < a, b < 1$, $X = 1/b$, $Y = 1/a$, and $m = \min(X, Y)$. Suppose Φ is homogeneous of degree one, continuous, and linear on each of $x \geq y$ and $y \geq x$. To prove (18), it suffices to check the origin, the seven other corners of $[0, X] \times [0, Y] \times [0, 1]$, and*

$$(m, m, 0), \quad (m, m, 1), \quad (0, 1, 1), \quad (X, 1, 1), \quad (1, 1, 1).$$

Duplicate points are counted once. All the nonzero points have $D = 1$.

Proof. Cut the box $D \leq 1$ by the planes $x = y$ and $y = z$. A vertex on neither plane is a box corner. A vertex on just one plane is a corner of the rectangle in which that plane meets the box. Vertices using both planes are the endpoints of the segment from $(0, 0, 0)$ to $(1, 1, 1)$. These possibilities give the listed points. On each cell, $N - \Phi(x, y) + \Phi(y, z)$ is the maximum of three affine functions, so its upper bound r follows from its values at the cell vertices. Apply this box inequality to a nonzero triple divided by $D > 0$, then use homogeneity. The all-zero triple is immediate. $\square$

6 A largest middle weight

This section proves $Q < rP$ when $a \geq 1$ in (15). We first give a binary estimate and then two local potentials.

Lemma 9. *For two integer support points with weights $g = (1, a)$, $k = (1, u)$, where $a \geq 1$, their complete convolution totals satisfy*

$$\frac{Q_2}{P_2} \leq \rho_a(u) := \max\left(\frac{1}{a}, u + \frac{a-1}{a^2}\right).$$

Consequently $Q < rP$ on every ordered three-point support with $g = (1, a, b)$ and $b \leq a/4$.

Proof. Split f into residue classes of the binary gap. The required local inequality is

$$\max(x, au) \leq \rho_a(u) \max(ax, y) + \lambda(x - y).$$

For $u \leq 1/a^2$, take $\rho_a(u) = 1/a$ and $\lambda = 0$. For $u \geq 1/a^2$, take $\rho_a(u) = u + (a-1)/a^2$ and $\lambda = 1/a - au \leq 0$. When $x > 0$, put $t = y/x$. The right side minus the left, divided by x , equals

$$\begin{cases} -\lambda t, & 0 \leq t \leq 1/(au), \\ 1 - t/a, & 1/(au) \leq t \leq a, \\ (u - 1/a^2)(t - a), & t \geq a. \end{cases}$$

These expressions are nonnegative; the breakpoint order follows from $a^2u \geq 1$. For $x = 0$ the difference is $(u - 1/a^2)y \geq 0$. Summing cancels the λ terms at both zero boundaries.

For three points let $W = \sum f$. Remove the last point from the numerator: $Q \leq Q_2 + bvW$, while $P \geq P_2$ and $P \geq aW$. Thus, when $b \leq a/4$,

$$\frac{Q}{P} \leq \max\left(\frac{1}{a}, u + \frac{a-1}{a^2}\right) + \frac{b}{a}v.$$

The first branch is at most $1 + v/4 < r$. The second is at most $u + (1 + v)/4 < r$, because $(a-1)/a^2 \leq 1/4$, equivalently $(a-2)^2 \geq 0$. Use Lemma 4 for both strict bounds. $\square$

Lemma 10. *On a progression, $Q < rP$ for every $0 < b \leq 1$ if either*

$$1 \leq a \leq 4/3, \quad \text{or} \quad 1 \leq a \leq 3/2 \text{ and } 0 < u \leq 2.$$

Proof. The range $br \geq 1$ is (19). Otherwise define

$$c = bv, \quad d_0 = \frac{1 - a(r - u)}{2 - a}, \quad d = \max(1/2, d_0), \quad \Phi(x, y) = x + d(y - x)_+.$$

We verify

$$\frac{1}{2} \leq d < 1, \quad d \leq \max(1/2, u), \quad a(r - u) + d(2 - a) \geq 1, \quad c < r - d. \quad (20)$$

For $a \leq 4/3$, the bound $r > u + 1/4$ gives $a(r - u) > a/4 \geq a - 1$, so $d_0 < 1$. The assertion $d_0 < u$ is equivalent to $ar > 1 + 2(a - 1)u$. For $u \leq 1$, the difference exceeds $(a - 1) + (2 - 7a/4)u$, which is nonnegative on $[0, 1]$ by its endpoint values $a - 1$ and $1 - 3a/4$. For $u \geq 1$, it exceeds $(2 - a)u + a/4 - 1 \geq 1 - 3a/4 \geq 0$.

For $a \leq 3/2$ and $u \leq 2$, (13) instead gives $a(r - u) > a/3 \geq a - 1$, again proving $d_0 < 1$. When $u \geq 1/2$, the difference $ar - 1 - 2(a - 1)u$ exceeds

$$(2a/3 - 1) + (2 - 4a/3)u = (1 - 2a/3)(2u - 1) \geq 0.$$

When $u \leq 1/2$, it exceeds $(a - 1) + (2 - 7a/4)u$; its endpoint values on $[0, 1/2]$ are $a - 1$ and $a/8$. Thus $d_0 < u$ in this range as well. The third condition in (20) follows from $d \geq d_0$. Finally, (14), $br < 1$, and $r > 1$ give

$$c < v/r < r - u, \quad c < v/r < r/2 < r - 1/2.$$

Together with $d \leq \max(1/2, u)$, these give the last condition.

We now check all three branches of (18). Since $0 \leq d \leq 1$, we have $\Phi(x, y) \geq x$ and $\Phi(y, z) \leq \max(y, z) \leq D$. As $r > 1$, the right side is at least x . For the other branches, $\Phi(x, y) \geq dy$ gives the lower bound

$$r \max(ay, z) + (d - 1)y - d(z - y)_+. \quad (21)$$

If $z \leq y$, this is $(ra + d - 1)y \geq [ra + d(2 - a) - 1]y \geq auy$. If $z \geq y$, the expression $r \max(ay, z) - dz$ is minimized at $z = ay$, so (21) is again at least $[ra + d(2 - a) - 1]y \geq auy$. For the last branch, when $z \geq y$ the bound is at least $(r - d)z + (2d - 1)y \geq (r - d)z \geq cz$. When $z \leq y$, use $ra + d - 1 \geq r - d > 0$ to reach the same bound. Since $c < r - d$, the padded triple $(0, 0, z)$ is strict for $z > 0$. Lemma 7 completes the proof. $\square$

Lemma 11. For $a \geq 1$ and $0 < b \leq 1$, the condition

$$ar - (a - b)u \geq 1 \quad (22)$$

implies $Q < rP$ on a progression.

Proof. Again assume $br < 1$. Put $p = a(1 - br)/(a - b)$ and $\Phi(x, y) = px$. Then $0 < p < 1$ and

$$\frac{1 - p}{b} + \frac{p}{a} = r.$$

The x branch follows from $(1 - p)x + py \leq [(1 - p)/b + p/a]D = rD$. For the middle branch, $rD + p(x - y) \geq (ra - p)y \geq auy$; the last step is precisely (22) after substitution of p . For the last branch, observe that

$$ar - b(a - b)v = [r - b(1 - b)v] + (a - 1)(r - bv) > 1.$$

Indeed $b(1 - b) \leq 1/4$, $r > 1 + v/4$, and $r > bv$. For fixed z , the expression $r \max(ay, z) - py$ is minimized at $y = z/a$, as $p > 0$ and $ra - p > 0$. Hence

$$rD + p(x - y) \geq \left(r - \frac{p}{a}\right)z = \frac{ar - 1}{a - b}z \geq bvez,$$

strictly if $z > 0$. This proves the local certificate. Its padded triple is strict, so Lemma 7 applies. $\square$

Proposition 12. *Theorem 1 holds on every progression whose middle g weight is at least both endpoint weights.*

Proof. Normalize as in (15), with $a \geq 1$. Lemma 10 covers $a \leq 4/3$; Lemma 9 covers $b \leq a/4$. Assume $a \geq 4/3$ and $b > a/4$. If $a \leq 3/2$ and $u \leq 2$, use Lemma 10. If $u \geq 2$, then

$$ar - (a - b)u = a(r - u) + bu > a/4 + (a/4)u \geq 3a/4 \geq 1.$$

If $a \geq 3/2$ and $u \leq 2$, then

$$ar - (a - b)u \geq a(r - 3u/4) > a(8/11 - u/44) \geq 15a/22 \geq 45/44 > 1.$$

Lemma 11 applies in the latter two cases. These cases cover all $a \geq 1$, including their shared boundaries. $\square$

7 Decreasing endpoint profiles

Proposition 13. *On a progression, $Q < rP$ for every profile $g = (1, a, b)$ with $0 < b \leq a < 1$.*

Proof. Assume $br < 1$, as the other range is (19). Put $\ell = a(1 - a) \in (0, 1/4]$. Taking the convex combination of (11) and $r > 1 + v/4$ with weights 4ℓ and $1 - 4\ell$ gives

$$r > 1 + \ell u + (1/4 - 3\ell/8)v.$$

Moreover,

$$b(1 - b - \ell) \leq \frac{(1 - \ell)^2}{4} \leq \frac{1}{4} - \frac{3\ell}{8}.$$

The difference in the second comparison is $\ell(1 - 2\ell)/8 \geq 0$. Consequently

$$r > 1 + \ell u + b(1 - b - \ell)v. \quad (23)$$

Define

$$\begin{aligned} p &= \frac{1 - br}{1 - b}, & s &= r - p = \frac{r - 1}{1 - b}, & \eta &= 1 - b - \ell, \\ \kappa &= \frac{a - b}{b(1 - a)}, & h &= \max\left(0, \frac{ab(1 - a)(u - s)}{\eta}\right), \\ A &= p + h, & B &= p + h/b, & e &= \max(0, p - \kappa h). \end{aligned}$$

All denominators are positive: $\eta \geq (1 - a)^2 > 0$. We have $0 < p < 1 < r$, $\kappa, h \geq 0$, and

$$s > bv, \quad h < b(s - bv) < bs, \quad B < r - bv, \quad B \geq A \geq p, \quad 0 \leq e \leq p < r. \quad (24)$$

The first assertion follows from $r > 1 + v/4$ and $b(1 - b) \leq 1/4$. For $h = 0$ the second is immediate. For $h > 0$, its first comparison is equivalent to $\ell(u - s) < \eta(s - bv)$, which is (23) because $\eta + \ell = 1 - b$ and $(1 - b)s = r - 1$. The remaining assertions follow from the definitions.

Four more coefficient comparisons will verify the local inequality:

$$(a - b)A + b(1 - a)e \geq a(1 - br), \quad (25)$$

$$e \geq \frac{1 - ar}{1 - a}, \quad (26)$$

$$r - e + \frac{B - A}{a} \geq \max(u, bv), \quad (27)$$

$$r - e + B - A \geq \max(au, bv). \quad (28)$$

For (25), substitute $p - \kappa h$ for e . The h terms cancel, giving equality; replacing this value by its nonnegative part can only increase the left side. For (26), direct subtraction gives

$$p - \kappa h - \frac{1 - ar}{1 - a} = \kappa(bs - h) \geq 0.$$

For (27), if $e = 0$ its left side is at least $r > \max(u, bv)$. If $e > 0$ and $h = 0$, it equals $s \geq u$ and $s > bv$. If $e > 0$ and $h > 0$, the identity

$$\kappa + \frac{1 - b}{ab} = \frac{\eta}{ab(1 - a)}$$

shows that the left side equals $s + h\eta/[ab(1 - a)] = u > s > bv$. For the au branch in (28), its left side is

$$a \left(r - e + \frac{B - A}{a} \right) + (1 - a)(r - e) \geq au.$$

For the bv branch it is at least $r - p = s > bv$.

Set

$$\Phi(x, y) = \begin{cases} Ax + ey, & x \geq y, \\ By + (A + e - B)x, & y \geq x. \end{cases} \quad (29)$$

The two formulas agree when $x = y$. Some coefficients may be negative; no sign condition on each individual coefficient is needed. Lemma 8 reduces the certificate to the origin and the following vertices. Write $\Delta\Phi = \Phi(x, y) - \Phi(y, z)$.

	(x, y, z)	$N(x, y, z)$	$\Delta\Phi$
1	$(1/b, 0, 0)$	$1/b$	A/b
2	$(0, 1/a, 0)$	u	$(B - A)/a$
3	$(0, 0, 1)$	bv	$-B$
4	$(1/b, 1/a, 0)$	$1/b$	$A/b + (e - A)/a$
5	$(1/b, 0, 1)$	$1/b$	$A/b - B$
6	$(0, 1/a, 1)$	$\max(u, bv)$	$(B - A)/a - e$
7	$(1/b, 1/a, 1)$	$1/b$	$A/b + (e - A)/a - e$
8	$(1/a, 1/a, 0)$	$\max(1/a, u)$	e/a
9	$(1/a, 1/a, 1)$	$\max(1/a, u)$	$e(1 - a)/a$
10	$(0, 1, 1)$	$\max(au, bv)$	$B - A - e$
11	$(1/b, 1, 1)$	$1/b$	$A(1/b - 1)$
12	$(1, 1, 1)$	$\max(1, au, bv)$	0

The numerator entries use $u, v < r < 1/b$, $a < 1$, and $bv < 1$. At $a = b$, repeated vertices in the table coincide, and all the comparisons below continue to hold.

Here are the checks of $N \leq r + \Delta\Phi$ for every row. Row 1 follows from $A \geq p \geq 1 - br$; row 2 from $B \geq A$ and $r > u$; row 3 is strict by $B < r - bv$. After multiplication by ab , row 4 requires $(a - b)A + be \geq a(1 - br)$, a consequence of (25) and $e \geq 0$. Row 5 is an equality, since $A - bB = (1 - b)p = 1 - br$. Rows 6 and 7 are (27) and (25), respectively. In rows 8 and 9 the u branch follows from $e \geq 0$ and $r > u$. For their $1/a$ branch, row 9 is exactly (26). For row 8 there is nothing to prove if $1 - ar \leq 0$; otherwise $e \geq (1 - ar)/(1 - a) \geq 1 - ar$ suffices. Row 10 is (28), and row 11 is $A \geq p$. Row 12 is strict because $r > \max(1, u, v)$, $a < 1$, and $b < 1$. The origin has numerator and potential difference zero.

The local certificate therefore holds everywhere. At $(0, 0, z)$ with $z > 0$ its right side is $(r - B)z > bvz$. Apply Lemma 7. $\square$

8 A smallest middle weight

Proposition 14. *On a progression, $Q < rP$ for every profile $g = (1, a, b)$ with $0 < a < b < 1$.*

Proof. Assume $br < 1$. Define

$$\begin{aligned} m &= 1/b, & n &= 1/a, & p &= \frac{m-r}{m-1}, \\ s &= r-u, & h &= \max\left(0, \frac{p-s}{m(m-1)}\right), & c &= bv, \\ A &= p+h, & B &= p+mh, & G &= B-A. \end{aligned}$$

Thus $1 < m < n$, $0 < p < 1$, $s > 0$, $h, G \geq 0$, and

$$(m-1)p = m-r, \quad mA - B = (m-1)p. \quad (30)$$

The binary scalar bounds and $b(1-b) \leq 1/4$ give

$$r-p = \frac{r-1}{1-b} > b \max(u, v) \geq c. \quad (31)$$

The definition of h gives

$$p - m(m-1)h = \min(p, s), \quad r + mG - p = r - \min(p, s) \geq u, \quad r + mG - p > c. \quad (32)$$

We also need the strict margin

$$B < r - c. \quad (33)$$

To prove it, note that $b(1-b)^2 \leq 4/27 < 5/32$; the first comparison follows from $4 - 27b(1-b)^2 = (3b-1)^2(4-3b) \geq 0$. Equation (11) therefore implies

$$r > 1 + b(1-b)u + b(1-b)^2v.$$

When $h = 0$, (33) is (31). When $h > 0$, $B = (mp - s)/(m-1)$ and the desired inequality is equivalent to

$$m(r-p) > u + (m-1)c.$$

On substituting $m = 1/b$ and (31)'s expression for $r-p$, this is exactly the preceding strict scalar bound.

Define the homogeneous potential

$$\Phi(x, y) = [p - (m-1)h]x + py + mh \max(x, y). \quad (34)$$

Its values at $(1, 0)$, $(0, 1)$, and $(1, 1)$ are respectively A , B , and $2p + h = A + p$. By Lemma 8, the complete nonzero vertex checks are as follows:

(x, y, z)	$N(x, y, z)$	$\Phi(x, y) - \Phi(y, z)$
$(m, 0, 0)$	m	mA
$(0, n, 0)$	u	nG
$(0, 0, 1)$	c	$-B$
$(m, n, 0)$	m	$mp + (n - m)G$
$(m, 0, 1)$	m	$(m - 1)p$
$(0, n, 1)$	$\max(u, c)$	$nG - p$
$(m, n, 1)$	m	$(m - 1)p + (n - m)G$
$(m, m, 0)$	m	mp
$(m, m, 1)$	m	$(m - 1)p$
$(0, 1, 1)$	$\max(au, c)$	$G - p$
$(m, 1, 1)$	m	$(m - 1)A$
$(1, 1, 1)$	$\max(1, au, c)$	0

The numerator values use $u, v < r < m$, $a < b$, and $c < 1$. For instance $aum = (a/b)u < u < m$ at the two interior $x = y$ vertices, while the middle numerator is $u < m$ at $y = n$.

Every row with numerator m has potential difference at least $(m - 1)p = m - r$, using $A \geq p$, $G \geq 0$, and $n > m$. At $(0, n, 0)$ use $r + nG \geq r > u$. At $(0, 0, 1)$ use (33). At $(0, n, 1)$ use

$$r + nG - p \geq r + mG - p \geq u, \quad r + nG - p > c$$

from (32). At $(0, 1, 1)$, (31) gives $r + G - p \geq r - p > b \max(u, v) \geq \max(au, c)$. At $(1, 1, 1)$ every numerator branch is less than r . The origin is immediate, completing all checks. The resulting local inequality is strict at $(0, 0, z)$ for $z > 0$, since its right side is $(r - B)z > cz$. Lemma 7 proves the proposition. $\square$

9 Completion of the proof

Proof of Theorem 1. For a non-arithmetic triple use Proposition 6. For a progression normalize as in (15). If $a \geq 1$, Proposition 12 applies. If $a < 1$ and $b = 1$, use the pointwise bound (19). The remaining profiles have $0 < a, b < 1$. If $b \leq a$, use Proposition 13; if $a < b$, use Proposition 14. This includes equal lower weights $a = b$ and every tie for the maximum. The residue decomposition and telescoping argument apply to arbitrary finite supports and amplitudes of f , with a strict inequality on each nonzero residue class. Thus $Q < rP$ in all cases. Since Q, P, r are positive and $r^4 = T$, raising to the fourth power proves (4). $\square$

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